

Midterm 1

STAT 109: Introductory Biostatistics — practice problems (binomial test and probability)

Midterm 1 — practice problems

The questions on the **in-class** portion of the midterm on **probability** and the **binomial test** will be drawn from these questions. Please try each question and compare your solutions with your classmates.

1. Dice

Suppose a pair of **fair six-sided dice** is rolled once.

- (a) List the **sample space** of all possible **sums** that can result from rolling the pair of dice once.
 - (b) How many ways can the sum of **7** be obtained when a pair of dice is rolled once?
 - (c) What is the **probability** of obtaining a sum of **7** if a fair pair of dice is rolled once?
 - (d) What is the **probability** of obtaining a sum of **7 or 11** if a fair pair of dice is rolled once?
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2. Screening test (conditional probability)

A university health center is evaluating a **screening test** for a rare viral infection among students. **4%** of students have the infection. If a student has the infection, the test is **positive 95%** of the time. If a student does **not** have the infection, the test is **negative 90%** of the time.

Let:

- D = student has the infection
- T^+ = positive test result

- (a) What is the probability that a **randomly selected** student tests **positive**?
 - (b) If a student tests **positive**, what is the probability that they **actually have** the infection?
 - (c) A student tests **negative**. What is the probability that they **truly do not** have the infection?
 - (d) Suppose **1,000** students are tested. How many students would you **expect** to test positive? Of those who test positive, how many would you **expect** to actually have the infection?
 - (e) A student says: “Since the test is **95% accurate**, if I test positive there’s a **95% chance** I’m infected.” Explain clearly why this statement is **incorrect**. Your explanation should reference the **base rate (prevalence)** of the disease.
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3. Binomial research questions (A–D) — preparation for the in-class analysis

For **each** of the **four** research questions (**A–D**) in the box below, be prepared to answer **any** of the follow-up questions **1–15**.

Note that an **observed value** of x for each question will be provided on the **in-class** exam for you to analyze and make a conclusion from based on what we expected based on an **appropriate plot** of x vs. $P(X = x)$.

i Research questions A–D

A. ChatGPT estimates that **47%** of adults in the US have **more than one pet**. Is the percentage of STAT 109 students who own **multiple pets different** than this figure? To find out, a statistics teacher plans to ask **40** students if they own more than one pet. The **number** of these students who **do** own more than one pet will be provided on the in-class midterm.

B. Is the percentage of college students with **colorful dyed hair less than 15%**? To find out, a statistics teacher plans to watch **20** students enter BSS and record x = number of them who have (in her opinion) colorful dyed hair. This **observed value** of x will be provided on the in-class midterm.

C. Is a pair of six-sided dice Rosanna purchased on eBay **more likely** to roll either a **7 or an 11** than a **fair** pair of dice? Suppose Rosanna is going to roll her pair of dice **35** times and record x , the **number of times** she gets the pair to sum to **either 7 or 11**. Her **observed value** of x will be provided on the in-class midterm.

D. Is the probability of a **correct forecast of rain** in Arcata **more than 0.9**? Suppose data from **14** days will be collected and x = “the number of times rain was **correctly** forecast” counted. This **observed value** of x will be provided on the in-class midterm.

For each of A–D, be able to answer:

1. State what each of the 4 conditions of the Binomial Random Process represents in the context of the research question.
2. Describe in a sentence what p represents in this context.
3. State the null and alternative hypotheses using the symbol p for the research question.
4. Let X be the number of successes in n trials. On the take-home exam, you created a graph of the probability of X assuming the null hypothesis is true, for values of X between 0 and n .

! Important

Note: this graph will be provided to you in-class during the exam.

Suppose that when data was collected, x out of n trials was observed. Add this data to the graph and decide whether to reject or fail to reject the null hypothesis.

Circle one: Reject the null hypothesis Fail to reject the null hypothesis

5. Based on your answer in part 4, write a one-sentence description of your conclusion based on the data about whether or not the null hypothesis is true.

6. Is a sample size of n large enough to use the Normal Approximation? Show your work in checking the technical condition for using the Normal Approximation to the Binomial.

7. Assume the sample size n is large enough for the Normal Approximation to be valid. Find the mean and standard deviation of X , the number of successes out of n trials assuming that the null hypothesis is true. Then use the Empirical Rule and the Normal Approximation to draw the distribution of X , the number of successes in n trials assuming the null hypothesis is true. Calculate and clearly label the bounds of the rejection region that excludes the middle 95% of the distribution.

8. Suppose that when data was collected, x out of n trials was observed. Add this data to the graph in 7 and decide whether to reject or fail to reject the null hypothesis.

Circle one: Reject the null hypothesis Fail to reject the null hypothesis

9. Based on your answer in part 8, write a one-sentence description of your conclusion based on the data about whether or not the null hypothesis is true.

10. Let X be the number of successes in n trials. On the take-home exam, you created a graph of the probability of X assuming the null hypothesis is true, for values of X between 0 and n .

! Important

Note: this graph will be provided to you in-class during the exam.

Suppose that when data was collected, x out of n trials was observed. Add this data to the graph and shade the area that represents the p -value for the alternative hypothesis.

11. Write an expression using the pdf $f(x) = P(X = x)$ of the Binomial random variable to compute the p -value for this data and research question with values for n , p , and x plugged in. Note: you don't need to evaluate this expression, just write it out.

12. Suppose the p -value for this data and research question is 0.0001. What do you conclude about the null hypothesis based on the size of this p -value?

Circle one: Reject the null hypothesis Fail to reject the null hypothesis

13. Suppose the p -value for this data and research question is 0.87. What do you conclude about the null hypothesis based on the size of this p -value?

Circle one: Reject the null hypothesis Fail to reject the null hypothesis

14. Suppose the p -value for this data and research question is 0.0001. Write a one-sentence interpretation of this p -value in the context of the data observed and the null hypothesis. 15. Suppose the p -value for this data and research question is 0.87. Write a one-sentence interpretation of this p -value in the context of the data observed and the null hypothesis.